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5. Sam carries out an experiment to determine the mass of a metal block using the principle of moments.

(a) State the principle of moments. [2]

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(b) The metal block is in the form of a cuboid as shown in **Diagram 1**. When setting up the experiment Sam accidentally knocks the metal block so that it tilts to the position shown in **Diagram 2**. The block then returns to the upright position as in **Diagram 1**.

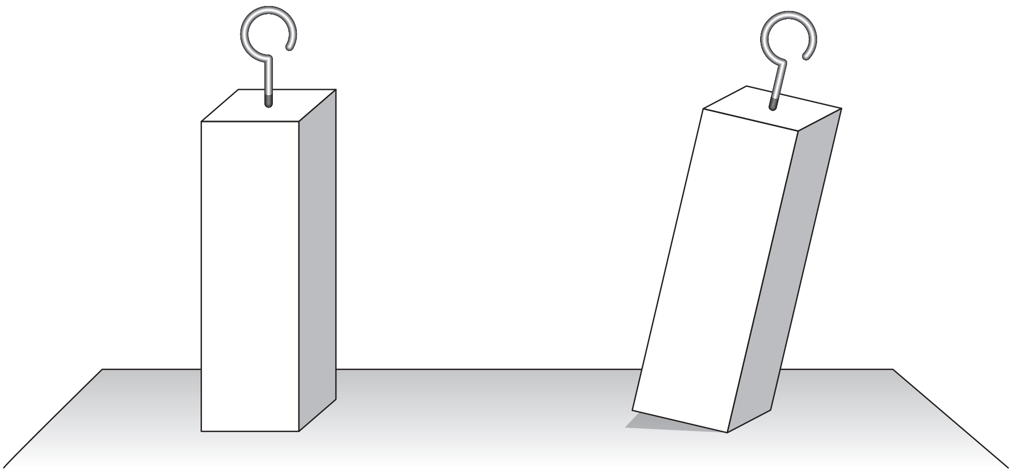

Diagram 1 **Diagram 2**

Explain, in terms of centre of gravity, why the block returns to the upright position. [2]

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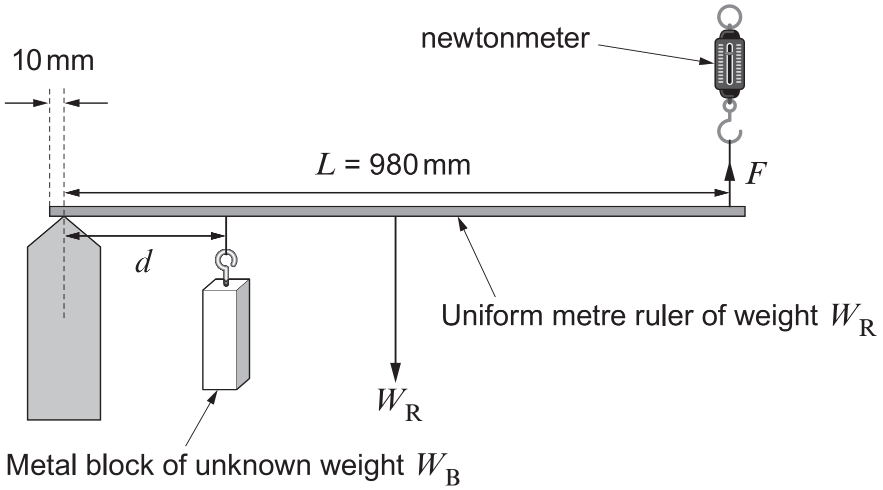
(c) Sam sets up the experiment as shown using a metre ruler.



Sam uses the newtonmeter to measure the force, F, needed to make the metre ruler horizontal for varying values of d. He carries out the experiment twice and determines a mean value for F for each of the distances d. Sam judges that the ruler is horizontal by eye before taking readings for F. The measurements are recorded in the table below.

Distance, d / mm	Force, F / N		
	Trial 1	Trial 2	Mean
100	2.8	2.6	
250	5.2	5.7	
400	8.8	8.5	
550	11.2	11.6	
700	14.6	14.5	
850	17.2	17.6	

(i) Complete the table. [1]

(ii) The following relationship is used to find the weight, W_B, of the block:

$$F = \frac{W_B d}{L} + \frac{W_R}{2}$$

By taking moments about the pivot, show how this relationship is obtained. [2]

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(iii) Use the grid to plot a graph of mean F (vertical axis) against d (in mm, on the horizontal axis) and draw a line of best fit. **Error bars should not be included.** [4]

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(iv) Use your graph to determine:

I. W_B, the weight of the metal block; [2]

II. W_R, the weight of the ruler. [2]

(d) The distances, d, were measured to a resolution of ±1 mm. Use data from the table of results to show that the percentage uncertainty in mean F is greater than the percentage uncertainty in d, when d = **100 mm**. [3]

(e) Suggest **one** change to his method which would improve the quality of Sam's measurements. [1]

19